ODE Model for Stemness

In the 2019 paper in the GitHub paper, Equation 11 defines stemness, σ as a function of APC, $a(t)$, and β-catenin, $\beta(t)$, given by

$$\sigma = \frac{\beta\sqrt{TCF^0}}{a\sqrt{K_{16}+\beta}},$$

where TCF^0 and K_{16} are given in Table 2 of the 2017 paper. I want you to formulate and write a Matlab code that outputs the result of an ODE for $d\sigma/dt$.

1.] Use the chain rule to find an expression for $d\sigma/dt$ by hand.

2.] Your code from Task 1 takes in a Wnt signal and outputs the solutions to da/dt (change in APC) and $d\beta/dt$ (change in β-catenin). So you already have these right-hand side functions defined. Further down in the same Matlab file, create a section for Stemness by defining the right-hand side for $d\sigma/dt$. Note that the previous model depends on Axin, but sigma doesn't have a dependence on Axin. Therefore, you will still need to consider Axin for the rates of change of APC and β-catenin and so the ODE for stemness is still dependent on three variables.

3.] The goal is to output a time-series of $\sigma(t)$ for any given Wnt signal.